

brian2-atlas

A Unified Intermediate Representation and Execution Architecture for Heterogeneous and Distributed Neural Simulation

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Next Brain

Execution semantics travel with the model

Order changes observations

A threshold, summed update or linked read can observe a different state after reordering.

Time extends across runs

Delayed events and mutable state must survive supported continuation boundaries.

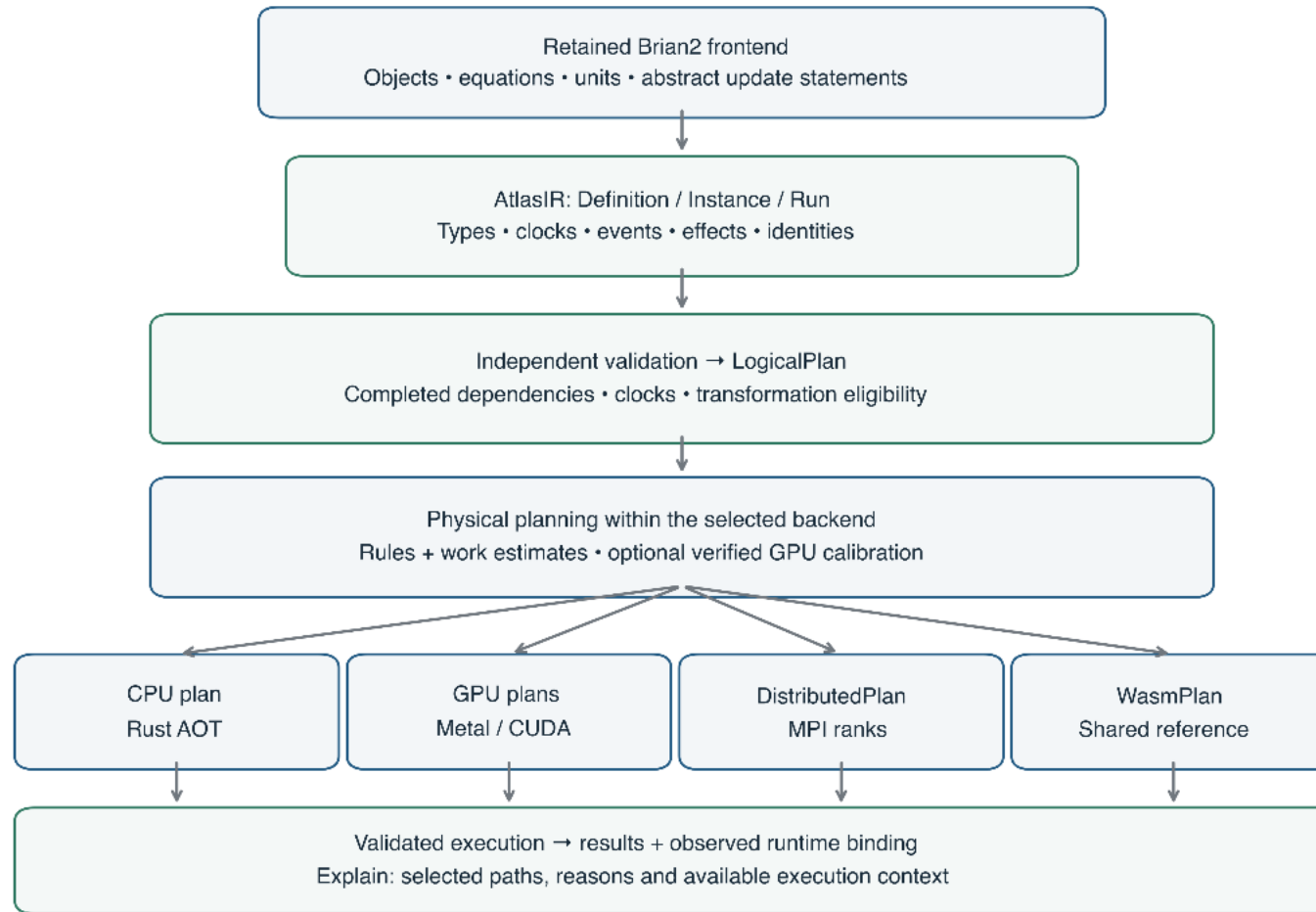
Partitions change ownership

Distributed execution must define where state lives and how spikes reach its owner.

Randomness needs identity

Global identities keep supported random streams tied to the model instance.

Atlas owns the downstream execution paths



One Device integration

Brian2 model preparation
AtlasIR validation
Target-specific execution

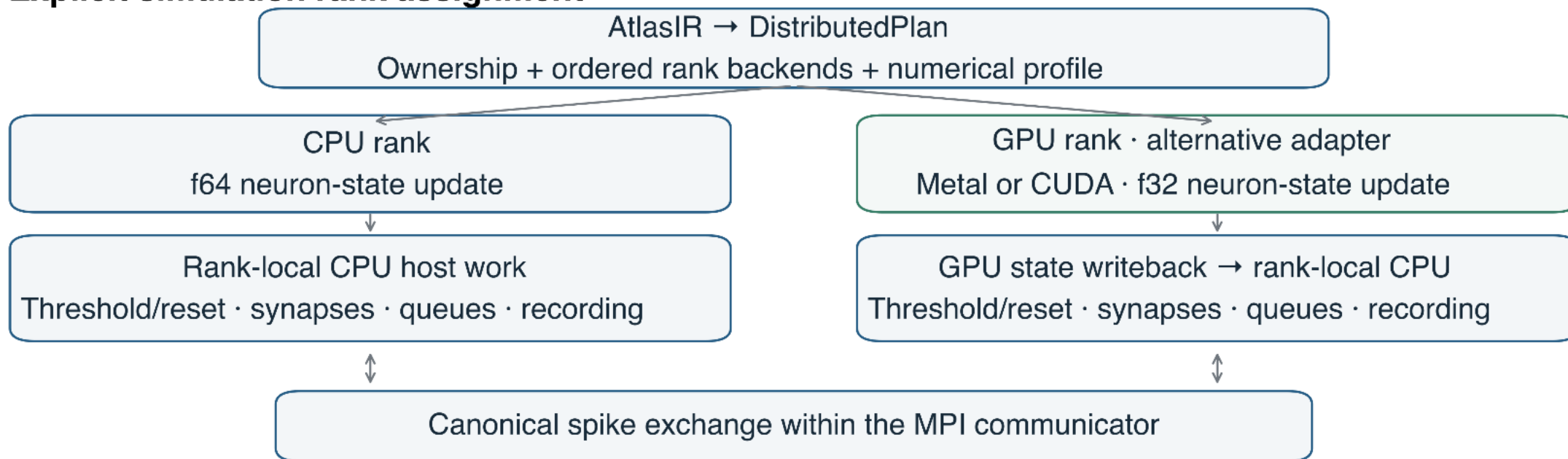
**No Brian2CUDA,
Brian2GeNN or GeNN
runtime dependency**

MPI distributes one model across machines

- Partitioned storage** Each rank holds local neuron state and incoming connections. Synapses belong to the rank owning the target neuron.
- Distributed build** Ranks build fixed-total connectivity from compact recipes, avoiding full recurrent-edge materialization in one Python process.
- Ordered events** Global neuron and edge identities survive partitioning. Ordered spike exchange drives local connections and delays.
- Capacity across hosts** Ranks can span hosts, distributing state and connection storage. Communication, global metadata and recording still limit scaling.

Heterogeneous execution with a backend per rank

a Explicit simulation rank assignment



Device selection and rank placement

Explicit CPU, Metal or CUDA per rank.
Ranks may span compatible hosts.

Qualification and deployment boundary

Multi-host CPU and local CPU+Metal are tested.
Compatible OS/CPU ABI and MPI are required.

The multi-area cortical model completed at full size

4.13M

neurons across 32 cortical areas

4 nodes / 32 ranks

100.5 s model time

10 h 55 min wall time

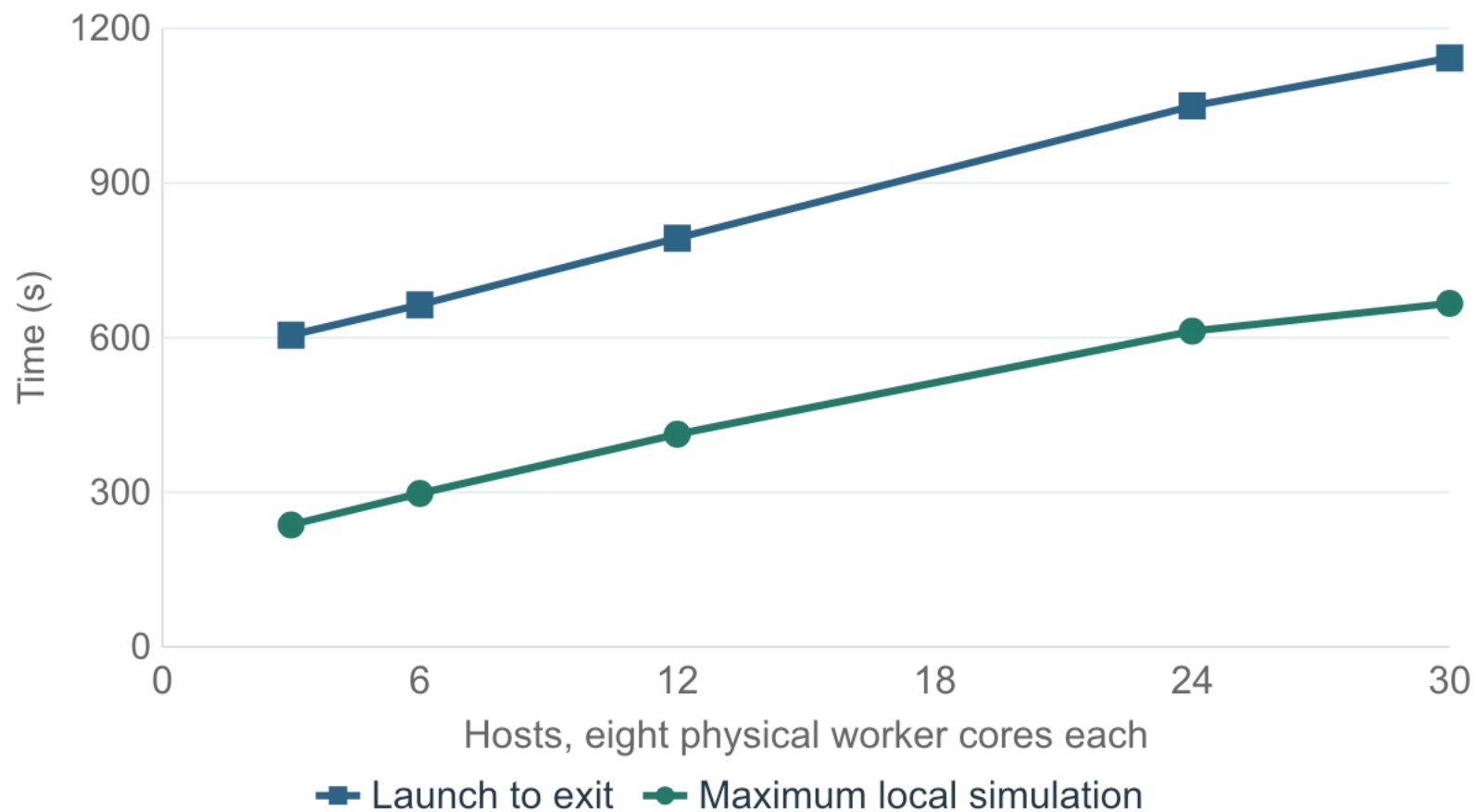
24.13B

recurrent synapses

A capacity and output-audit result

Biological equivalence and matched
NEST performance remain unestablished.

Weak scaling reaches 860 million neurons



860B

connections at 30 hosts

100 ms

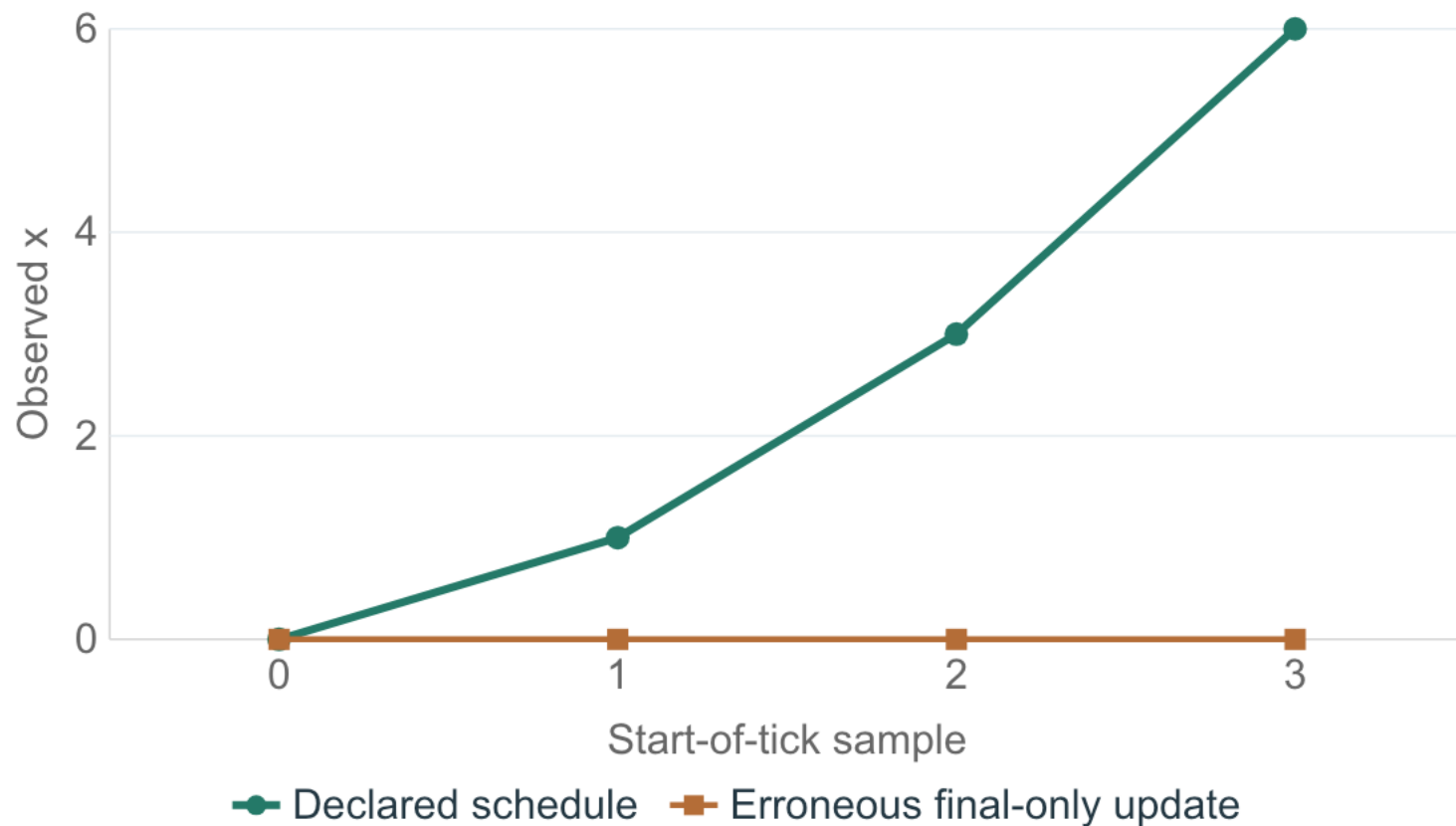
model time
Synthetic recurrent E/I

AtlasIR separates definition, instance and run

Layer	What it identifies
Definition	Populations, update program, schedules, clocks and numerical contracts
Instance	Initial state, parameters, topology, pending events and seed
Run	Absolute start time, duration and clock tick intervals

Independent validation checks types, dimensions, events and declared effects.

A linked read makes update order observable



[0, 1, 3, 6]

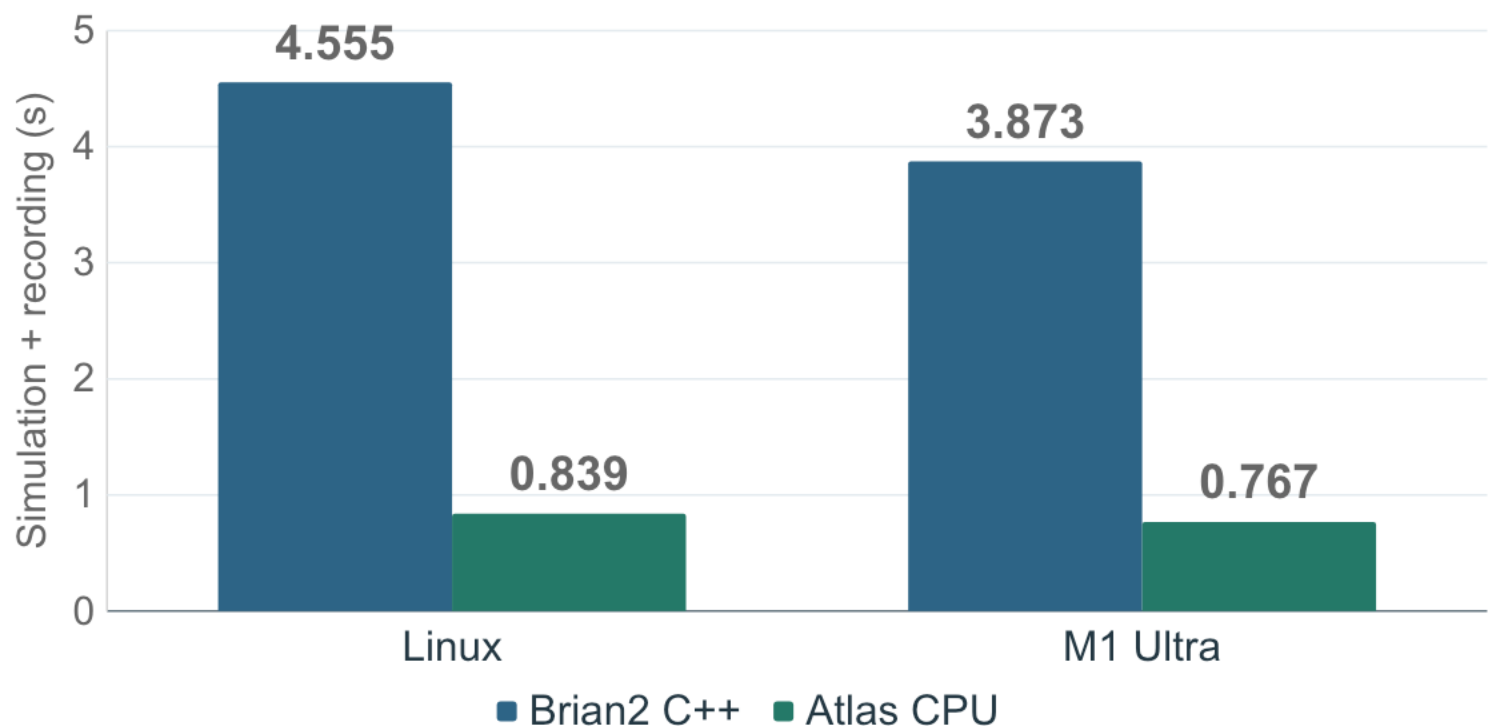
Completed read sets expose intermediate consumers and constrain optimization.

Each result has a defined measurement scope

Evidence	Question answered	Examples
Conformance	Do checked outputs agree?	CPU arrays, event order, continuation
Performance	How long is a stated interval?	CPU replay, GPU host-array replay
Capacity	Did this configuration complete?	PD14, multi-area cortex, 860M
Functionality	Which combinations passed?	Browser, native training, mixed MPI

Qualification applies to the tested combination of model, target and protocol.

CPU full-connectome replay improves on both hosts



5.43×

Linux median-time ratio

5.05×

M1 Ultra median-time ratio

Best measured configurations per backend. Thread counts differ.

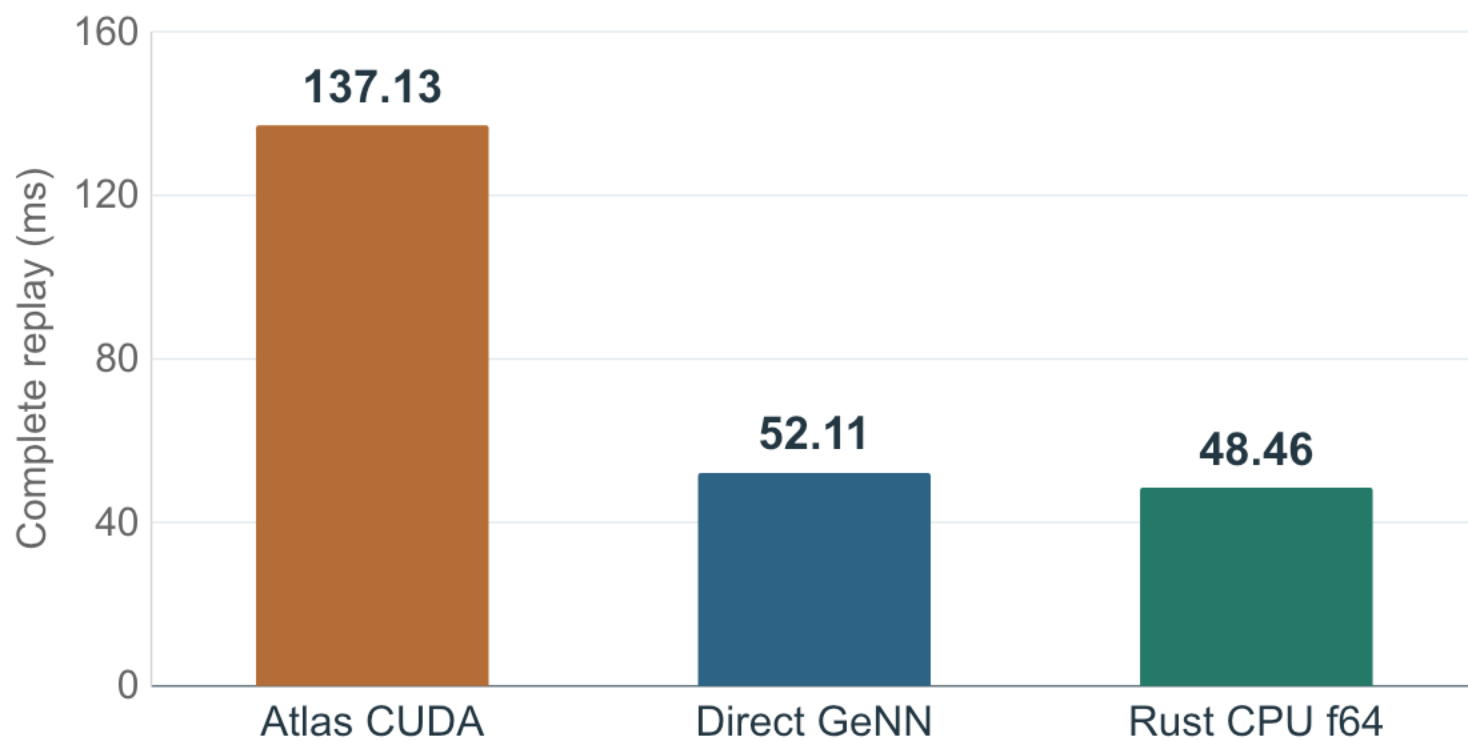
GPU replay depends on the workload

Host / accelerator	Atlas GPU (ms)	Rust f64 control (ms)
M3 / Metal	67.19	85.73
L4 / CUDA	34.82	216.02
A100 / CUDA	32.58	181.80

The replay includes reset, execution and complete host arrays.

GPU uses f32. The reference control uses f64 on each corresponding host.

Recurrent CUBA exposes a slower CUDA case



4,096 neurons
131,072 connections
2,048 ticks

Passing a numerical gate
does not ensure a faster
physical plan.

298.9 million synapses on a 16 GB laptop

298.9M

synapses

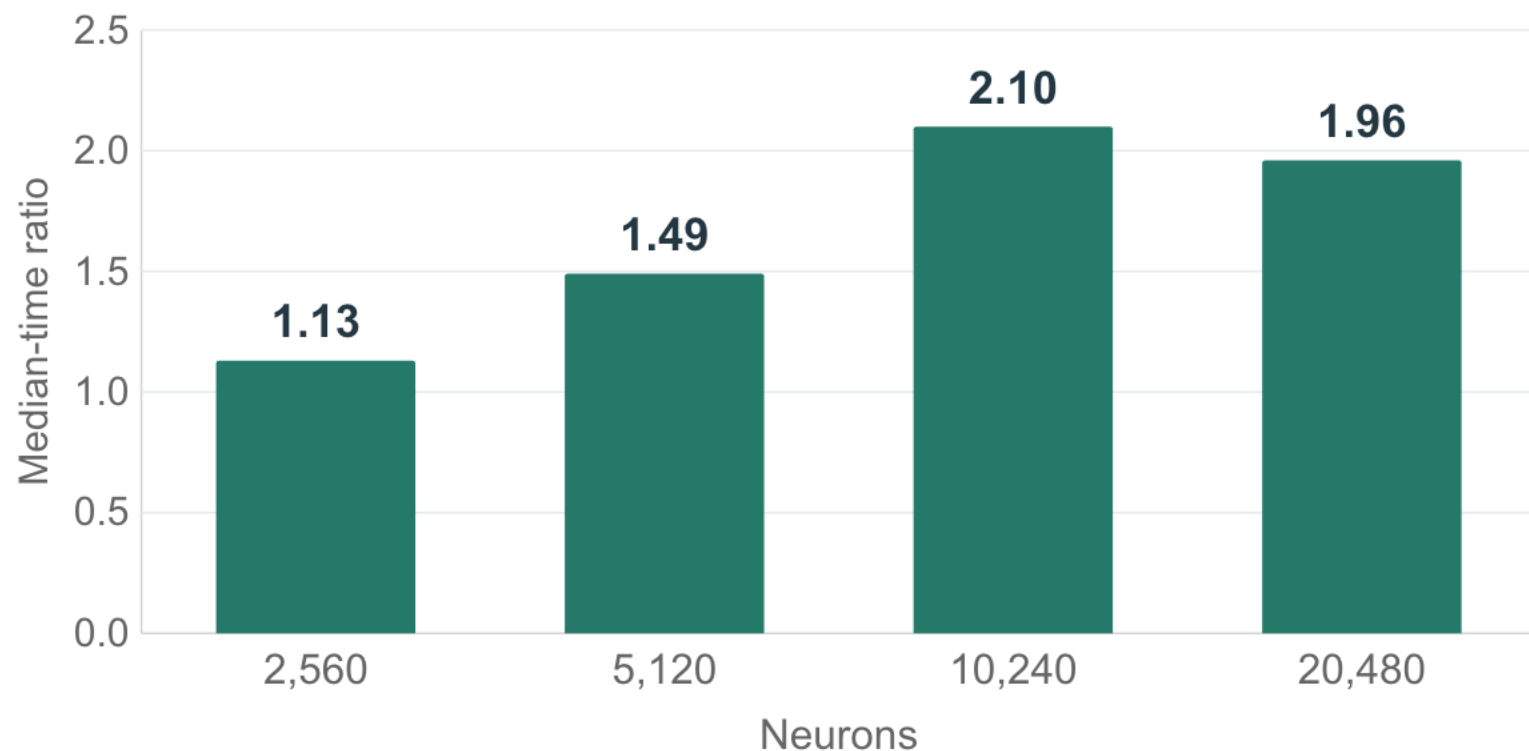
10 s

model time completed

Measured interval / resource	Atlas
Simulation body	104.009 s
Frontend + build + run + load	125.586 s
Native-child peak	6.314 GB
Separate frontend peak	0.319 GB

C++ preparation exceeded its controlled budget before simulation.

Explicit NMDA workloads extend the CPU evidence

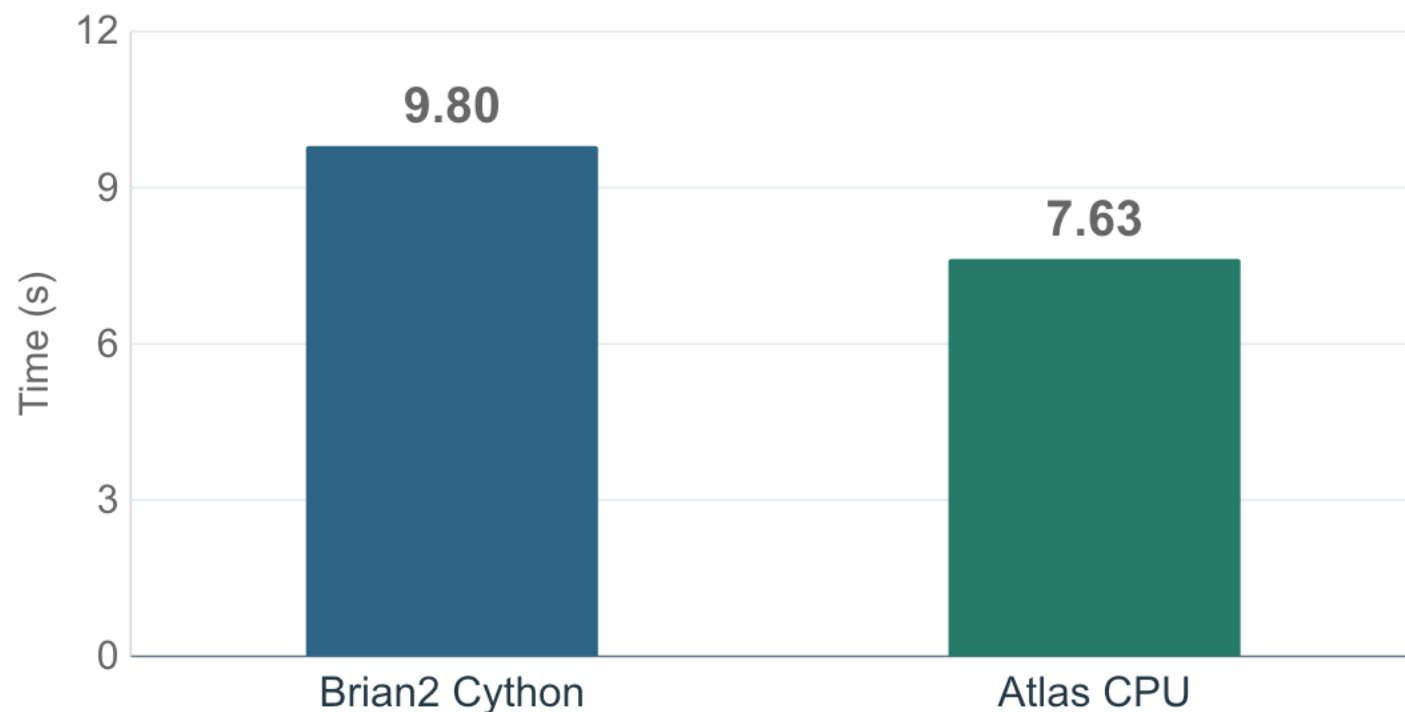


Up to 2.10×

Matched eight-core
compiled-region timing

5–8% higher
sampled native RSS

Endpoint reuse improves the dendritic network



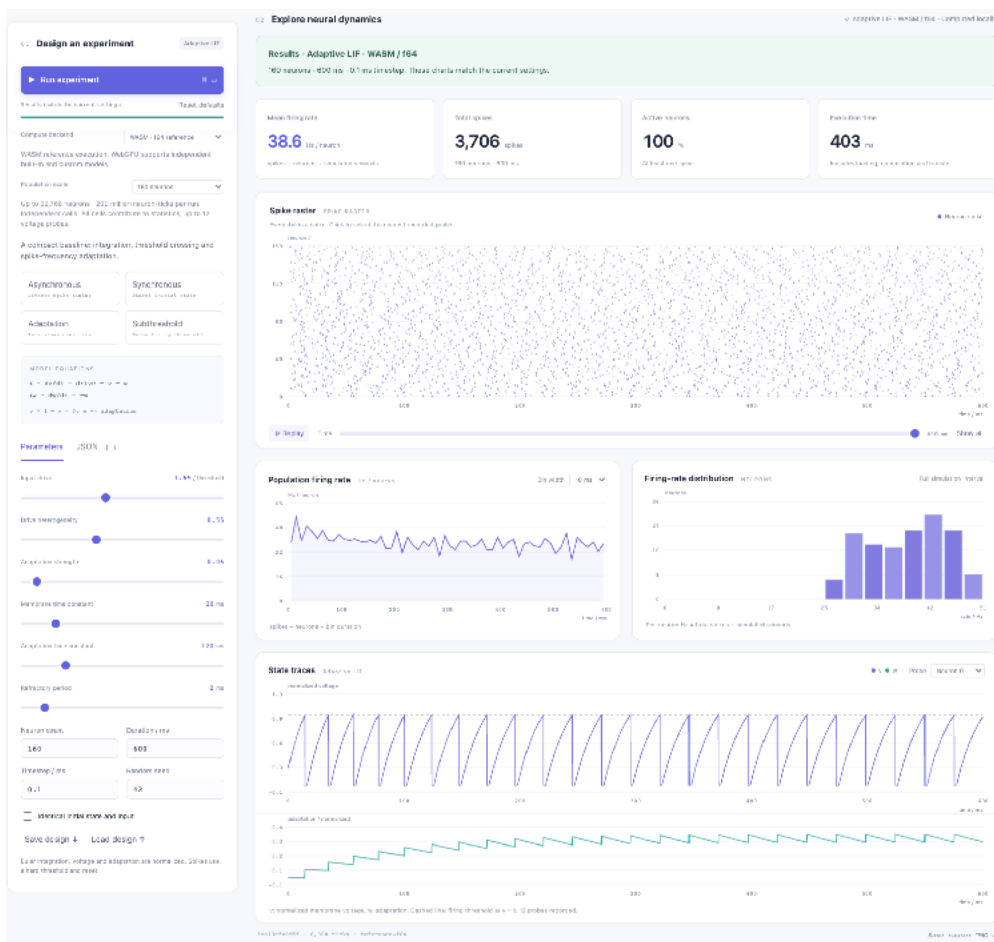
1.28×

Cython / Atlas median ratio

**34.0% less Atlas
runtime**

In the separate cache-policy
ablation

Browser execution has distinct profiles



Generic Wasm

Validated bundles and a shared reference runtime power local experiments.

Separate full-connectome AOT

139,255 neurons and 15.09M weighted edges. Thirteen fixed-input CPU/Wasm checks.

WebGPU currently admits a narrower independent-cell f32 profile.

Training and mixed MPI have bounded acceptance

Path	Evidence in the preprint	Scope boundary
Native training	709 passed on Modal L4	One GPU, including two shared-GPU ranks
CPU + Metal simulation	Local rank ownership and dispatch checks	Neuron-update offload on one host
CUDA MPI simulation offload	Source generation and adapter implemented	No NVIDIA execution qualification
Cross-host mixed devices or multi-GPU training	Further qualification needed	No corresponding performance or scaling claim

Surrogate gradients and continuation contracts do not establish dataset accuracy.

The current evidence has clear limits

Eligibility remains target-specific

A valid AtlasIR document does not guarantee support on every execution target.

Capacity is one acceptance level

Completion at large scale does not establish biological equivalence or a NEST speed ratio.

Performance varies by workload

Precision, dispatch and lifecycle costs can reverse the ordering of backends.

Scaling costs remain global

Global metadata, communication and recording still grow with network size.

Public implementation, evidence and preprint

One model, partitioned across machines

AtlasIR constrains execution semantics

Preprint v2

doi.org/10.5281/zenodo.23273257

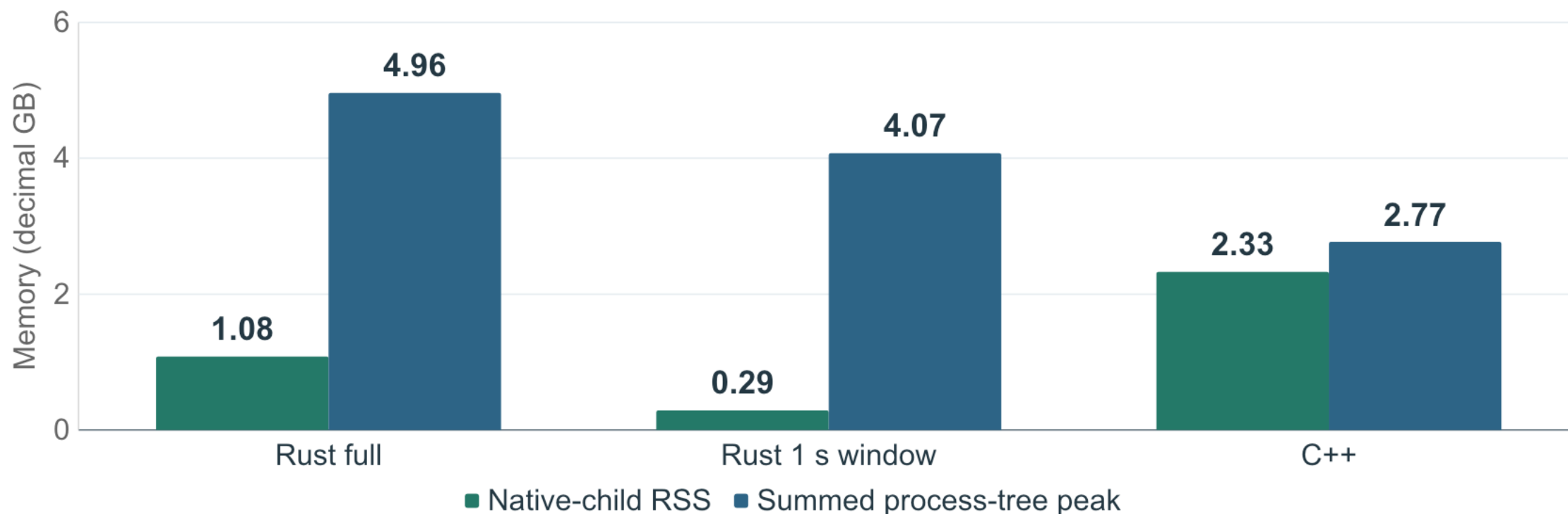
Software v0.1.0

github.com/RockLi/brian2-atlas
doi.org/10.5281/zenodo.23269098

Evidence archive

github.com/RockLi/brian2-atlas-preprint
doi.org/10.5281/zenodo.23269145

Backup: native and process-tree memory differ



A rolling native buffer does not bound all memory in the workflow.

Backup: verified GPU calibration has a cost

Wide case	Baseline (ms)	Selected (ms)	Calibration (s)
L4, prefix + bitset	122.26	79.86	24.36
A100, prefix + bitset	154.33	91.75	23.99
M3, baseline retained	1,313.67	1,313.67	16.87

Selection-round reductions: 34.7% on L4 and 40.6% on A100.

Calibration can cost much more than a single replay.